Bi-Lingual Audio Detection for Sentiment Analysis

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Abstract—This study proposes a hybrid deep learning framework in the field of sentiment analysis of bilingual (Hindi-English) audio recordings, acknowledging the problems caused by code-switching in mixed language conversation. This model based on Bi Directional Long short-term memory (BiLSTM) and long short-term memory (LSTM) networks captures temporal and contextual patterns for the classification of emotions, such as happiness, fear, sadness, and neutrality. Results demonstrate the capability of the model to capture emotional nuances in code-switched audio; these abilities may enable further applications in multilingual environments, including virtual assistants and automated customer support systems. The research establishes a fundamental platform for addressing code-switching problems in audio sentiment analysis.

Keywords—Bilingual Sentiment Analysis, Hindi-English Sentiment Detection, Audio Sentiment Analysis, Code-Switching, Emotion Detection, Language Identification.

I. Introduction

With the rise of voice based technology, virtual assistants, and automated customer service solutions, the necessity of capturing user sentiment through their speech rather than textual based sentiment analysis has become strikingly clear. While text sentiment analysis is more often used in applications such as customer care, social media tracking, and mental well-being examination, audio sentiment analysis makes it much easier to assess user sentiments in real time. The above mentioned migration (text to Audio based sentiment

Analysis) results in huge challenges, especially in multilingual settings, where speakers constantly alternate languages during a single conversation (known as code-switching). Particularly in countries such as India, where there is large number of languages spoken together, code-switching complicates sentiment analysis because traditional monolingual models are incapable of detecting emotions in mixed-language inputs. Hindi and English are often mixed in conversations that happen daily in India, and fast changes and switches between languages within a single audio stream pose major complications to the models of sentiment recognition. Most of the sentiment analysis techniques are monolingual, and code-switched bio-lingual data tend to decline their performance and reliability when applied to real-world situations.

The current study elaborates a strong framework for sentiment analysis over code-switched, multilingual audio data, granting credence to the Hindi-English language pair. Audio sentiment analysis consists extracting relevant acoustic properties from an audio stream including pitch, and chroma obtained from an audio signal using Mel-frequency Cepstral Coefficients (MFCCs).

A hybrid model of deep learning with CNNs and Bi LSTM is proposed to accommodate the complex temporal patterns in bilingual audio data. Here, features are extracted using CNN. Since CNN preserves the temporal context for all time steps. This architecture classifies the processed code-switched audio signals into four emotions: happiness, fear, sadness, and neutral.

As far as sentiment analysis on multilingual text is concerned, it has made remarkable strides forward; however, a technologically sophisticated audio-based model capable of processing bilingual, code-switched audio streams in a highly efficient manner still eludes us. The existing sentiment classification methods fail to notice or identify emotion indicators in multilingual audio data simply because of the resilience requirements necessary for using switching languages. This performance gap thus prevents the practical deployment of sentiment-based applications like virtual assistants and automated customer support systems in multilingual spaces.

Contribution of proposed work:

- It lays foundation for Multi-lingual Sentiment Analysis using Multimodal hybrid and deep learning method.
- Creation of labeled audio dataset.
- Increase in performance over the existing methods.
- Handling the code switching issue

II. Related Work

The field of Bi lingual or Multi lingual models for sentiment analysis has been explored in vast amount of ways yet failed to handle the issues like mixed language, audio based real time analysis etc. It can be approved using deep learning and multi model architecture [1]. At the same time, a comparison of deep learning algorithms for multi lingual sentiment analysis (MLSA), focusing on the difficulties like lack of dataset pre-processed multilingual data, to obtain better MLSA accuracy [3],[5].

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The integration of multimodal approaches has emerged as a promising strategy for sentiment analysis. Experiment shows that there is a potential for future studies to explore multimodal fusion methods [4]. However, the trend is toward fusion techniques with greater quality and real-time information access [13],[14]. The sentiment analysis problems in code-mixed and sparse languages have also been explored [25]. Simultaneously, one stated the importance of artificial sentiment assets and sophisticated deep learning algorithms for the proper administration of informal and code-mixed languages [17].

Machine learning (ML) approaches still hold an important position in multilingual sentiment analysis as its learning architecture are capable of recognizing the pattern enabling the handling of heterogeneous dataset[7]. Furthermore, the relevance of using fully convolutional neural networks pointed out the necessity for better datasets for sentiment classification in non-fixed-length audio streams [11].

Deep learning models have also been demonstrated to be applicable in certain applications [18]. In the same way, the survey showed cross-lingual sentiment analysis, which addressed pre-trained models and pointed out issues, like fine-grained analysis in low-resource languages [19]. To overcome the problem of code-switching, use of bilingual deep neural networks for automatic code-switching detection in Frisian speech and showed that multilingual training methods are needed [6]. High words error rates as a result of code-switch are still a critical constraint creating stresses for the need for balanced data sets [10].

The combination of auditory, textual, and visual modalities has emerged as a reoccurring method in sentiment analysis. Development of a multimodal sentiment analysis system that achieved state-of-the-art performance using Facial features and transformer based learning [13]. Likewise, proof advocated for incorporating advanced facial recognition algorithms to improve sentiment detection accuracy in news videos, while exploring the application of reinforcement learning to enhance multimodal fusion [14],[15].

Aspect-based sentiment classification has also been a focal point. A multi-layer network for cross-lingual aspect-based sentiment analysis obtained a significant increase in precision, recall, and F1 score by analyzing co-occurrence and dependency relations [16].

Cross-lingual approaches have shown promise in addressing under-resourced languages. For example demonstration of the feasibility of cross-lingual phoneme mapping using multilingual acoustic models achieving high word accuracy rates [9]. Moreover, a comprehensive review of sentiment analysis in Indian indigenous languages and argued for the utilization of state-of-the-art models, namely BERT for code-mixed language classification [12].

TABLE I. LITERATURE SURVEY OF AUDIO BASED SENTIMENT ANALYSIS

S. No.	Title/Author	Year	Future Scope	Limitations	Outcome
1.	Joint sentiment analysis of lyrics and audio in music by "Technische Hochschule N¨urnberg Georg Simon Ohm" [16]	2024	Multimodal models have been improved using Russell's circumplex model and huge, high-quality bimodal datasets for sophisticated cross-modal sentiment classification.	Challenges in emotion taxonomy standardization include subjective perception, small datasets, and modality differences in emotion labels.	60/40 audio-text weighting was found to be successful for emotion prediction, highlighting the need for enhanced multi modal techniques to improve music-based sentiment analysis.
2.	A Systematic Literature Review of Modalities, Trends, and Limitations in Emotion Recognition, Affective Computing, and Sentiment Analysis by "Rosa A. García-Hernández, Huizilopoztli Luna-García" [30]	2024	Non-intrusive emotion recognition technologies are being developed, as is multi modal fusion of physiological and physical information and interdisciplinary research integration.	Standardization challenges include interpersonal signal variability, difficult physiological data analysis, and a lack of large representative datasets for emotion recognition.	EEG-based emotion studies are growing, there is a need for non-intrusive emotion detection approaches, and multi-signal fusion has the potential to improve emotion recognition systems.
3.	Challenges and future in deep learning for sentiment analysis: a comprehensive review and a proposed novel hybrid approach by "Md. Shofqul Islam1,2 · Muhammad Nomani Kabir.eta" [31]	2024	Merging knowledge-driven strategies, addressing various formats and real-time data, and investigating novel approaches for sentiment detection over the social web.	Text sentiment analysis presents challenges with coherence, negation, intensifiers, semantic meaning, and dealing with complicated sentence patterns.	The CRDC model demonstrated great accuracy across several datasets, highlighting the efficiency of deep learning; capsule-based RNN techniques produced superior accuracy of up to 98.02%
4.	Emotion recognition and artificial intelligence: A systematic review (2014–2023) and research recommendations by "Smith K. Khare,Victoria Blanes-Vida.eta"mean [19]	2024	Create public datasets, incorporate feature/data fusion, investigate hybrid DL algorithms (BI LSTM, LSTM, transformers), and use federated meta-learning to achieve robust emotion recognition.	Limited research due to a lack of publicly available datasets; difficulties in obtaining model explain-ability, uncertainty quantification, and trustworthiness.	SVM has been found as useful for ML-based emotion recognition, however DL models (particularly BI LSTM) are gaining popularity because to their capacity to extract deep features, hence improving system performance in a wide range of applications.
5.	Advanced Emotion Recognition and Regulation Utilizing Deep Learning Techniques by Yoheswari S [26]	2024	Improve system adaptability to complex emotional states, integrate multi modal information (visual, aural, and physiological), and investigate unsupervised learning to lessen dependency on labeled	Ethical considerations, such as privacy and data security; difficulty in recording delicate emotions and contextual conditions.	The deep learning framework reached 92% accuracy, exceeding typical machine learning approaches in real-time applications, contributions include high precision emotion

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			data for real-world adaptability.		recognition and an adaptive emotion management system.
6.	Investigating fairness in machine learning-based audio sentiment analysis by Yang Liu, Sophina Luitel. e.al [34]	2024	Future research should explore the impact of demographic factors beyond gender, such as race, age, and ethnicity, on audio sentiment analysis. Additionally, validating findings with diverse datasets and investigating non-binary gender representations are crucial.	The study was limited to binary gender categories due to dataset constraints and may not generalize to non-binary identities. Biases related to other demographic factors were not addressed.	Gender-specific models outperformed the gender-agnostic model in audio sentiment analysis, with accuracies of 78% for female-specific and 74% for male-specific models compared to 66% for the gender-agnostic model. An ensemble approach improved overall accuracy to 76%.
7.	Sentimental analysis of audio-based customer reviews without textual conversion by Sumana Maradithaya1, Anantshesh Katti [33]	2023	Automation of the prepossessing pipeline, flexibility to varied methodologies, and medical applications for mental health therapy.	Existing systems rely on textual audio conversion, which takes time; sarcasm in audio can confound sentiment analysis.	Sentiment analysis accuracy improved by 10% over previous models, audio feedback was used well in CRM systems, and there is potential for medical applications.
8.	Multilingual Sentiment Analysis Using Deep Learning: Survey by Uswa Ihsan, Humaira Ashraf.eta [3]	2023	Future research should focus on techniques for low-resource languages, domain adaptation, cross-lingual transfer learning, curated multilingual datasets, and language-specific preprocessing for enhanced sentiment analysis accuracy.	Multilingual sentiment analysis faces challenges like data scarcity in low-resource languages, domain adaptation, and cross-lingual transfer, requiring curated datasets and language-specific techniques for improved accuracy.	Expected outcomes include accurate multilingual sentiment analysis, enhanced decision-making, robust models for diverse applications, and new research opportunities, advancing the field of multilingual sentiment analysis.
9.	Attention-based Spatialized Word Embedding Bi-LSTM Model for Sentiment Analysis by Kun Zhu and Nur Hana Samsudin [8]	2023	Extend the model to more sentiment analysis scenarios and integrate features from sound and video using multi modal joint representation for improved sentiment text classification.	The limitations of this paper include the model's focus on movie reviews only, potential challenges in extending the approach to other domains, and the need for integration with multi modal data for broader sentiment analysis.	Improved sentiment analysis across various contexts, enhanced accuracy through multi modal data integration, and better classification performance in diverse applications.
10.	Multi modal Abusive Language Detection and Sentiment Analysis using Transformer based architecture by Abhinaba Bala, Parameswari Krishnamuthy [12]	2023	Future research should expand datasets to address resource limitations, explore advanced multi modal fusion techniques, incorporate contextual and temporal information, and address class imbalance issues.	Current challenges include limited dataset resources, the need for advanced integration methods, and addressing class imbalance, with a focus on Dravidian language nuances.	The proposed framework effectively integrates visual, auditory, and textual features for detecting abusive content and sentiment analysis, showing promising results in Dravidian language videos.

Primary Objective based on research:

- Multimodal Approach – Development of multimodal system for deeper analysis of data for enhanced learning.
- Handling Bi-lingual Data specifically Hindi & English.
- Handling Code Switching issues
- Creation of Custom Audio labeled dataset

III. Methodology

This section describes the preparation of the data set utilized in this study as well as the methodologies used to analyze bilingual audio samples for sentiment analysis. The data set was carefully chosen to contain a wide range of audio sources, ensuring comprehensive coverage of Hindi and English feelings voiced in a variety of settings.

A. Dataset Collection

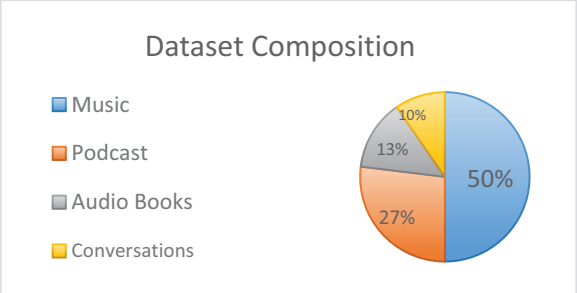

Fig. 1. Dataset Composition

As shown in Fig.1 the collection is made up of audio recordings from various platforms and formats that are intended to represent real-world multilingual communication circumstances. The main sources of audio data are:

Music: Selected Hindi and English songs with emotive lyrics offer valuable sentiment data for analysis.

Podcasts: A collection of bilingual podcasts with conversational and monolingual content.

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Audio books: Excerpts from popular Hindi and English audio books provide emotional expression and storytelling opportunities.

Voice notes: Collected voice messages from several folks to capture informal and spontaneous communication.

Conversation Recordings: Recorded talks between bilingual speakers can help analyze natural code-switching and emotional expression in real-time encounters

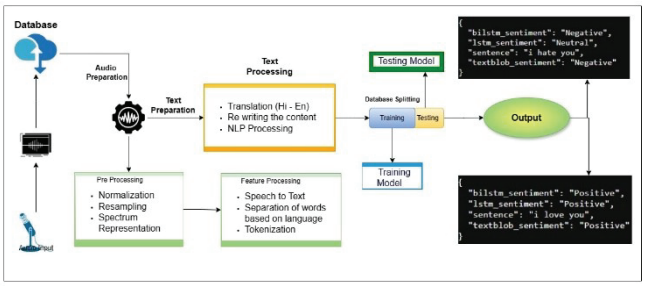

Fig. 2. Diagram of Methodology

B. Data Annotation and Preprocessing

Fig. 1 shows the percentage distribution of entire dataset for example music, podcast, audiobooks etc that hold significant percentage like 50%, 27%, 13% respectively.

After collecting audio samples, the dataset was annotated to characterize the emotions exhibited in each recording. The following steps were followed for annotation and preprocessing:

Emotion Labeling: Each audio clip was carefully examined to determine and categorize the predominant emotional state (happy, fearful, sad, or neutral). The labeling method was carried out by native Hindi and English speakers to ensure linguistic and cultural accuracy in sentiment categorization. Emotion labels, designated as E={e1,e2,e3,e4}. The technique used the OpenSmile toolkit for feature extraction and emotion recognition, as well as Pandas to efficiently organize and annotate the classified data.

Transcription: To help with feature extraction and analysis, a subset of the audio samples was transcribed using pre-trained Whisper models from OpenAI to capture spoken words and expressions, ensuring a robust and versatile sentiment analysis system. Librosa and Speech Recognition libraries supported audio processing during transcription. This transcription enabled cross-verification of sentiment and contributed to a better comprehension of the context in each audio.

Translation and Rewriting: For Multilingual Content: Translation of the Hindi text was done in English using machine language subtitling and Google translator in order to better align semantic interpretation.

Rewriting text from the perspective of better preserving linguistic nuances such as idiomatic expressions and culture-specific references, and clarity and coherence. The spaCy library was used to manage linguistic complexities and maintain accurate contextual rewriting.

C. Data Preprocessing:

Noise Reduction: Data were treated to remove background noise and improve clarity. Libraries used in this are librosa, scipy and BERT.

Normalization: the training data was standardized to reduce the variance in the quality of the data. Text data cleaning, removal of punctuations, special characters, reduction of space.

Resampling: All audio data was resampled to ensure similar sampling rates, which ensured compatibility across different processing pipes and preserved consistent frequency representation.

Feature Extraction: Key audio properties were extracted from each sample and used as inputs for machine learning models. These characteristics include spatial features , pitch, tone, and intensity.

Segmentation: Data were split into smaller and manageable parts. In order to be properly evaluated in sentiment, this segmentation was very necessary, ensuring constant language and dependable emotion classification.

D. Methodology of Sentiment Analysis

Sentiment analysis was carried out utilizing a hybrid model that combined (BI LSTMs) and Long Short-Term Memory (LSTM). This method enables the model to detect both spatial information from audio spectrograms and temporal relationships in sequential data. The methodological flow for sentiment analysis is shown in Fig.2. The sentiment analysis technique, which used Tensorflow and Keras, consisted of evaluating features derived from preprocessed dataset and classifying them based on sentiment thresholds.

Model Training: To enable effective model training and evaluation, the dataset was partitioned into three sets: training, validation, and testing. The training set taught the model to recognize patterns associated with various sentiments, while the validation set helped fine-tune hyper parameters. The data splits into testing set and training set was done to validate the generalization performance of the model on unseen data. Libraries such as scikit-learn and TensorFlow aided in data splitting, model training, hyperparameter tuning, and performance evaluation on the test set.

Metrics Evaluation: To measure the performance of the LSTM, and BiLSTM models, the following metrics were computed:

- Precision: Fraction of correctly predicted positive instances,
- Recall: Fraction of actual cases of positive instances that are correctly classified,
- F1-Score: A single performance metric that balances other two-metrics outcome of precision and recall.

Furthermore, confusion matrices were generated to give insights into the class-wise accuracy of each of the models across the four sentiment categories.

E. Module Workflow

The sentiment analysis module's workflow as shown in Fig.2. was designed to acquire, process, and evaluate audio data in real time. The following stages define the module workflow:

Start- Real-Time Audio Capture: The system continuously listens for audio input from numerous sources, allowing for the recording of live encounters.

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Audio Analysis and Transcription: Once audio has been taken, it is analyzed to discover essential elements before being transcribed into text format. Audio analysis is commonly handled by librosa for feature extraction, while transcription is done using speech recognition libraries such as speech recognition, Google Speech API and open AI whisper.

Transcribed Text and Rewriting: The transcribed text can be rewritten or cleaned to improve clarity and coherence, allowing for more accurate sentiment identification. Libraries like nltk, spacy, and re (regular expressions) are used for text cleaning and preprocessing.

Text Analysis Using NLP Processing: Natural language processing techniques like tokenization, lemmatization, and sentiment lexicons are applied to the cleaned text to examine the linguistic features and patterns present, assisting in comprehending the context of the sentiments expressed. Libraries such as nltk, spacy are employed for NLP tasks, helping to understand the context of the sentiments expressed.

Sentiment Analysis: The hybrid BILSTM-LSTM model analyzes the processed text for sentiment, categorizing it as happy, fearful, sad, or neutral. The models are built and trained using tensorflow, keras, and scikit-learn for model development and evaluation. It also uses mixout regularization and attention mechanism to avoid over fitting of the data that can result in poor accuracy.

Result and Evaluation: Finally, the sentiment analysis findings are gathered, providing the identified sentiments, as well as their probabilities and evaluation metrics. The results are presented using matplotlib and seaborn for visualization, and pandas was used for data handling.

By following this structured module work-flow as depicted in Fig.2, the study ensures a systematic method to recording and analyzing multilingual audio data, improving the accuracy and reliability of sentiment recognition in both Hindi and English.

IV. Experimental Findings And Analysis

In this section, we show our experimental results and compare the performance of various algorithms for sentiment analysis in multilingual audio detection. Long Short-Term Memory (LSTM) & Bi-directional LSTM (BiLSTM), were examined on F1-score, Precision, and Recall metrics to determine their efficacy in emotion recognition inside Hindi-English multilingual audio snippets.

As it shown in Table 2, the LSTM has the highest F1-score, Precision, and Recall values of 0.9711, 0.9913, and 0.9911, respectively, followed closely by the BiLSTM model, which produces marginally lower scores whilst displaying comparable performance in the handling of sequential data. The BiLSTM model was similarly trained for 5 epochs with identical parameters as the LSTM model. Both the LSTM-based models performed well for sequential based pattern recognition and learning.

TABLE II. Comparative Analysis Of Both Separate Model

S.No.	Algorithm	F1_Score	Precision	Recall
1.	Lstm	0.9710766	0.9913265	0.9910714
2.	Bilstm	0.9710670	0.9912891	0.9910714

The comparative analysis with other models as showed in Table 3 concludes that our approach of hybrid framework shows better performance than other existing models. This finding indicates that combining LSTM-based architectures is critical for obtaining high accuracy in bilingual sentiment analysis, particularly for audio data including complicated, code-switched linguistic patterns.

TABLE III. Comparison With Different Model's Performance

Model	F1 Score
XGBOOST	56%
Sentiment analysis on speaker specific [27]	89%
Audio Spatial Information based analysis [27]	75%
Our Proposed Model	95%

V. Limitations, Future Scope And Conclusion

A. Limitations

There is a bright scope ahead for bilingual audio sentiment analysis with this approach.

Multiple limitations are also associated as following:

1) Dataset Constraints:

- Although rich in recording yet does not cover all the required emotional states.
- Dialect Variation is high leading to reduction in performance.

2) Code Mixed based challenges:

- Model is effective for pure hindi and English conversations but lacks explicit support for hinglish training.
- Lack of explicitly addressed and casual bilingual interactions.

3) Audio Feature extraction based challenges:

- Variation in sound quality and background affects the sentiment analysis by introducing bias.
- Speakers specific characteristics also effects the overall analysis.

VI. Future scope

This present research provides possibilities for further studies. The main goal is to enrich the data by adding further emotions such as fear and surprise. Such an extension will benefit the model greatly and allow it to be used in better applications. Another topic regarding future work alternating with this would be how to consider emotions expressed in 'Hinglish.' It may involve building something completely new or upgrading the already-established framework in more capable ways of understanding emotions for mixed languages. It would be great if studies used more advanced natural language processing tools and models, such as transformers and attention-mechanisms, to analyze contextual changes in sentiment's realizations. Other suggestions would be to bring in another dimension to this research that is multi modal sentiment analysis where images or videos along with texts are analyzed to draw sentiment from the conversations.

VII. Conclusion

In conclusion, this work shows that deep learning models, notably LSTM and BiLSTM, are successful at capturing the complex patterns required for sentiment analysis in Hindi-

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English bilingual audio data. The proposed approach outperformed the existing implemented approaches by atleast 7% and also enables the real time analysis that increases applications of the model. Future research could look into other advancements, such as merging multi modal data sources and increasing datasets, to improve sentiment detection capabilities in a variety of linguistic settings.

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